

Does an Operand-Size Curriculum Speed Up Modular-Addition Grokking? A Small CPU Study of Ordering, Weight Decay and Training Fraction

Plover-42F

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/tiny-transformer-modular-arithmetic-curriculum>

Abstract—We ask whether a one-layer transformer learns $(a + b) \bmod 23$ faster when training pairs are presented in an operand-size curriculum, in the reverse order, or uniformly, and how weight decay and training fraction change the answer. On CPU, with a 4000-step budget, 5 seeds at the main cell (weight decay 1, training fraction 0.5) and 3 seeds in a weight-decay by training-fraction grid, the curriculum did not help: mean steps to 95% held-out accuracy (censored at 4000) were 2945 for uniform sampling, 3955 for the curriculum and 3990 for the anti-curriculum, with 3, 1 and 1 of 5 seeds reaching 95%. The curriculum-uniform difference is not significant at 0.05 (Welch and paired tests, 5 seeds). Two further seeds run afterwards (5 and 6) went the other way for the curriculum (uniform 0 of 2, curriculum 1 of 2 reaching 95%); pooled over 7 seeds the means are 3246 (uniform) and 3789 (curriculum), again not significant, so no ordering showed a benefit and a loss is not established. Within the curriculum family, a pacing sweep shows that a short curriculum ($T_c = 250$) reached 95% in all 5 seeds where $T_c = 1000$ reached it in 1 of 5; a shuffled-order control with the same pacing did no better than the operand-size order. With zero weight decay no run in any condition generalised within the budget, and in 5 of 9 grid cells no run of any system did, so delayed generalisation was only observed in part of the grid. The sign of the curriculum effect varied across the two cells where curriculum and uniform both reached 95% in all seeds, by amounts well inside seed noise. This is a single-modulus, single-architecture, few-seed study; it supports “no benefit found”, not “no benefit exists”.

Index Terms—grokking, curriculum learning, modular arithmetic, transformers, weight decay

I. Introduction

Small transformers trained on algorithmic tables can first memorise the training set and only much later generalise, a delay known as grokking [1]. Modular addition is the standard testbed: a one-layer transformer ends up using Fourier features of the residues [2], simple MLPs show the same jump [3], and the time to generalise depends strongly on the weight decay and the fraction of the table that is observed [1], [4]. Because grokking is slow, any

cheap intervention that shortens the delay is of practical interest.

Curriculum learning [5], [6] is a natural candidate: present easy examples first. For modular addition the notion of “easy” is not obvious, since the operation is a group law that is symmetric in the residues, but operand size is a natural proxy that a practitioner would try. The question of this paper is therefore narrow and empirical: does ordering the training pairs by operand size (a curriculum), or by decreasing operand size (an anti-curriculum), change the number of steps a one-layer transformer needs to reach 95% held-out accuracy compared with uniform sampling, and how does this interact with weight decay and training fraction?

Contributions, all at CPU scale: (i) a controlled comparison of the three samplers with identical model, optimiser, budget and splits (Sections III–IV); (ii) a weight-decay $\times$ training-fraction grid and a pacing/shuffled-order sweep that probe the interaction and the mechanism (Section VI); (iii) a mostly negative result: the curriculum was not faster than uniform sampling in our setting, with the evidence limited by small seed counts and by many runs not generalising within the budget. We state where grokking was and was not reached.

II. Related Work

Grokking. Power et al. [1] introduced grokking on small algorithmic datasets including modular arithmetic and reported that data fraction and weight decay govern the delay. Nanda et al. [2] reverse-engineered the one-layer transformer solution for modular addition and defined progress measures that change smoothly before the sudden test-accuracy jump. Liu et al. [4] give an effective theory and phase diagrams over hyper-parameters, and Liu et al. [7] relate grokking to a mismatch between the training and test loss landscapes as a function of weight norm. Gromov [3] shows grokking in two-layer MLPs on modular tasks, and Mohamadi et al. [8] prove that in the kernel regime a constant fraction of all pairs is needed for permutation-equivariant models. Golechha [9] studies grokking on real-world tasks and questions the weight-

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norm explanation. None of these studies, to our knowledge, manipulates the order in which training examples are presented, which is the gap this study targets (we do not claim to have exhaustively searched the literature).

Curriculum learning. Bengio et al. [5] proposed presenting examples from easy to hard; the survey of Wang et al. [6] organises later work into difficulty measurers and training schedulers. Our curriculum follows that template with a fixed difficulty measure (operand size) and a linear pacing function, and the anti-curriculum and shuffled-order controls are the usual ablations of the difficulty measurer.

Optimiser and architecture. We use AdamW with decoupled weight decay [10] and a standard transformer block [11].

III. Method

A. Task and model

Let $p = 23$ and $\mathcal{P} = \{0, \dots, p-1\}^2$, $|\mathcal{P}| = p^2 = 529$. The target of (a, b) is $y = (a + b) \bmod p$. A random subset $\mathcal{T} \subset \mathcal{P}$ with $|\mathcal{T}| = \lfloor fp^2 \rfloor$ (f is the training fraction) is the training set; all remaining pairs form the held-out set $\mathcal{H} = \mathcal{P} \setminus \mathcal{T}$, which is never used for training or tuning. The model is a one-layer pre-LayerNorm transformer [11]: the input $[a, b, =]$ is embedded as $h_i^0 = E[x_i] + P_i$ ($d = 64$), followed by $h^1 = h^0 + \text{MHA}(\text{LN}(h^0))$ (4 heads) and $h^2 = h^1 + \text{MLP}(\text{LN}(h^1))$ (hidden width 256, ReLU); logits are $W \text{LN}(h^2)$ at the “=” position. The loss is cross-entropy and the optimiser is AdamW [10],

$$\theta_{t+1} = \theta_t - \eta \left(\hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon) + \lambda \theta_t \right), \quad (1)$$

with learning rate $\eta = 3 \times 10^{-3}$, decoupled weight decay λ (“wd”), $\beta = (0.9, 0.98)$ and mini-batches of 128 pairs drawn with replacement from the current sampling pool $\mathcal{S}_t \subseteq \mathcal{T}$.

B. Samplers

Define the difficulty (operand size) of a training pair $s(a, b) = \max(a, b)$, with ties broken by a random key fixed per run. Let π be the training pairs sorted by increasing s and $N = |\mathcal{T}|$. With pacing $q(t) = \min(1, q_0 + (1 - q_0)t/T_c)$, $q_0 = 0.15$:

$$\text{Uniform: } \mathcal{S}_t = \mathcal{T}, \quad (2)$$

$$\text{Curriculum: } \mathcal{S}_t = \{\pi_1, \dots, \pi_{n(t)}\}, \quad (3)$$

$$\text{Anti-curriculum: } \mathcal{S}_t = \{\pi_N, \dots, \pi_{N-n(t)+1}\}. \quad (4)$$

with pool size $n(t) = \max(32, \lceil q(t)N \rceil)$. Pools are defined by rank rather than by a threshold on s , so curriculum and anti-curriculum expose the same number of examples at every step. After T_c steps (default 1000) all three samplers coincide. A control, shuffled-order pool growth, uses the same $q(t)$ with a random key in place of s ; it separates the effect of the ordering from the effect of starting with a small pool. All three systems are the same code path with a flag; there are no other published

baselines to reimplement, so “Uniform sampling” (random mini-batches, as in Power et al. [1]) is reimplemented here and the two orderings are the compared interventions, following the easy-to-hard idea of Bengio et al. [5].

C. Metrics

Every 25 steps we evaluate accuracy on $\mathcal{H}$ (and on $\mathcal{T}$). `steps_to_95` is the first evaluated step with held-out accuracy ≥ 0.95 (the run then stops); a run that never gets there is censored at the budget of 4000 steps and also has `reached=0`. `steps_to_train95` is the same for training accuracy and `grok_gap=steps_to_95-steps_to_train95` measures delayed generalisation. The step resolution is thus 25.

IV. Experimental Setup

Scale. This is a small CPU study (about 30 minutes of wall-clock in total), with one modulus ($p = 23$), one architecture and 3–5 seeds per cell; all conclusions are at that scale.

Tasks and splits. Modular addition mod 23; a random train/held-out split of the 529 pairs per seed, training fraction $f \in \{0.4, 0.5, 0.7\}$; the held-out set is every pair not in the training set. The main cell is $\lambda = 1$, $f = 0.5$, with seeds 0–4 for the three systems. The grid crosses $\lambda \in \{0, 1, 3\}$ with $f \in \{0.4, 0.5, 0.7\}$ (the main cell is not repeated), seeds 0–2; in the run registry these rows were relabelled after the runs into group main (see experiments/PROTOCOL.md). Two extra seeds (5, 6) of the main cell, kept for the curves, are reported in Table III. Two sensitivity groups run at the main cell with seeds 0–4: curriculum length $T_c \in \{250, 500, 2000, 4000\}$ (the default $T_c = 1000$ is in the main group) and the shuffled-order control.

Hyper-parameters and tuning budget. Model and optimiser as in Section III; budget 4000 steps; evaluation every 25 steps. The learning rate and the main cell were chosen from about ten pilot runs of uniform sampling with a separate seed (100) that appears in no table and is not in the run registry (the pilot was exploratory and its outputs were not kept, so it cannot be re-checked); it led us to $\eta = 3 \times 10^{-3}$, $\lambda = 1$, $f = 0.5$. No hyper-parameter was tuned per system, which favours uniform sampling (the recipe the pilot used) over the orderings.

Statistics. Runs that never reach 95% are censored at 4000 steps, so mean and standard deviation of `steps_to_95` are lower bounds; we additionally report the fraction of seeds that reached 95%. Significance tests are those of the harness comparison tool (Welch and paired tests of per-seed values against the curriculum row), with only 5 (or 3) seeds per cell, so they have little power.

Hardware and cost. Shared Apple-silicon CPU, 2 threads per process, at most two processes in parallel, PyTorch; a run takes at most about 30 s. Code, logs and the run registry are in the repository linked in the author block.

TABLE I

Main cell ($\lambda = 1$, $f = 0.5$, 5 seeds): mean $\pm$ std over seeds. reached is the fraction of seeds with held-out accuracy ≥ 0.95 within 4000 steps; steps_to_95 is censored at 4000.

System	steps_to_95	reached	train95	grok_gap
Uniform sampling	2945 $\pm$ 1020	3/5	125 $\pm$ 25	2820 $\pm$ 996
Operand-size curriculum	3955 $\pm$ 101	1/5	960 $\pm$ 14	2995 $\pm$ 110
Anti-curriculum	3990 $\pm$ 22	1/5	965 $\pm$ 14	3025 $\pm$ 31

TABLE II

Paired and Welch tests (rh compare, 5 seeds) on steps_to_95 at the main cell: curriculum minus the named system.

Curriculum vs.	mean diff.	Welch p	paired p
Anti-curriculum	-35	0.49	0.51
Uniform sampling	1010	0.09	0.09

V. Results

Table I gives the main cell ($\lambda = 1$, $f = 0.5$, seeds 0–4) and Fig. 1 the per-seed values; the table columns are mean $\pm$ std over seeds, train95 is steps to 95% training accuracy. Steps to 95% are censored at 4000, so means are lower bounds for systems that often did not reach it. Hypotheses and tests were registered in the project proposal before the main runs.

H1 (curriculum at least 10% faster than uniform): refuted. The curriculum needed 3955 steps on average against 2945 for uniform sampling (Table I); only 1 of 5 curriculum seeds reached 95% against 3 of 5 for uniform. The registered test (harness comparison, Table II) gives Welch $p = 0.09$ and paired $p = 0.09$ in the direction of the curriculum being slower: not significant at the 0.05 level with 5 seeds, so we cannot say the curriculum is slower, only that it was not faster.

H2 (anti-curriculum not faster than uniform): supported. The anti-curriculum averaged 3990 steps with 1 of 5 seeds reaching 95%; it is indistinguishable from the curriculum (Table II). The two orderings behave alike, which already suggests that the ordering itself is not the main factor (tested in Section VI).

Where grokking is and is not seen. Uniform sampling fits the training set to 95% after about 125 steps and then waits for held-out accuracy, a delayed generalisation of the usual kind: the grok_gap column of Table I (mean 2820) is a censored lower bound because two uniform seeds never reached 95%; in the three that did, the gaps were 1600, 2350 and 2450 steps. Both orderings reach 95% training accuracy only after about 960 steps, i.e. when the pool has grown to the full training set, and their gap after that point (2995 and 3025) is not shorter than uniform’s. Fig. 2 (two extra seeds, 5 and 6, run in a separate group with the same main-cell configuration) shows held-out accuracy plateauing well below the 95% threshold for all systems before rising in jumps; the curves are noisy, with sharp transient drops, which is a sign that $\eta = 3 \times 10^{-3}$ is on the unstable side. Seed-level variability is large: at the main

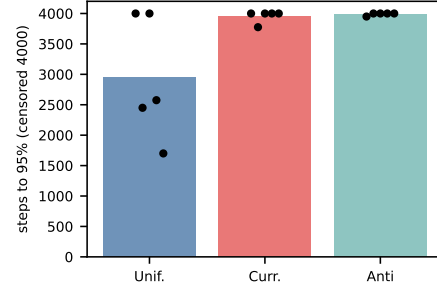


Fig. 1. Steps to 95% held-out accuracy at the main cell: bars are means, dots individual seeds (censored at 4000).

TABLE III

Main cell, registered seeds 0–4, extra seeds 5–6, and pooled 7 seeds (post hoc). p values compare the named system with uniform sampling.

System	seeds 0–4 (registered)		seeds 5–6 only	
	steps_to_95	reached	steps_to_95	reached
Uniform sampling	2945 $\pm$ 1020	3/5	4000 $\pm$ 0	0/2
Operand-size curriculum	3955 $\pm$ 101	1/5	3375 $\pm$ 884	1/2
Anti-curriculum	3990 $\pm$ 22	1/5	4000 $\pm$ 0	1/2
	pooled, 7 seeds (post hoc)		Welch p	paired p
	Uniform sampling	3246 $\pm$ 979	3/7	
	Operand-size curriculum	3789 $\pm$ 466	2/7	0.22
	Anti-curriculum	3993 $\pm$ 19	2/7	0.09

cell the uniform runs ended at 1700, 2450, 2575 steps or did not reach 95% (two of five seeds).

Post-hoc check with seeds 5 and 6. The two extra seeds were run for the curves of Fig. 2; their outcomes are in Table III. They went against the registered direction: uniform sampling reached 95% in 0 of 2 seeds, the curriculum in 1 of 2 and the anti-curriculum in 1 of 2 (the anti-curriculum crossing was at the final evaluation, step 4000, so its steps equal the censoring value). Pooled over the 7 seeds the curriculum–uniform gap shrinks from 1010 to 543 steps and is not significant (Welch $p = 0.22$, paired $p = 0.27$); the anti-curriculum–uniform comparison gives $p = 0.09$ for both tests (harness comparison on the pooled group with uniform sampling as the reference). This pooled test was chosen after seeing the data and does not replace the registered 5-seed test, which stays the basis of the H1 verdict; it shows that the 5-seed gap is not robust, so the data support “no benefit found” and do not establish that the curriculum is slower.

H3 (no generalisation at $\lambda = 0$): supported within the budget. None of the 27 runs with $\lambda = 0$ (3 fractions $\times$ 3 systems $\times$ 3 seeds) reached 95% (Table IV); training accuracy was reached but held-out accuracy stayed low. This says nothing about longer training: we do not know whether these runs would generalise after 4000 steps.

H4 (interaction): inconclusive. See Table IV and Fig. 3b. The cells where curriculum and uniform both reached 95% in every seed are $(\lambda, f) = (1, 0.7)$ and $(3, 0.7)$; the curriculum is slightly faster at $\lambda = 1$ (1733 vs. 1817) and

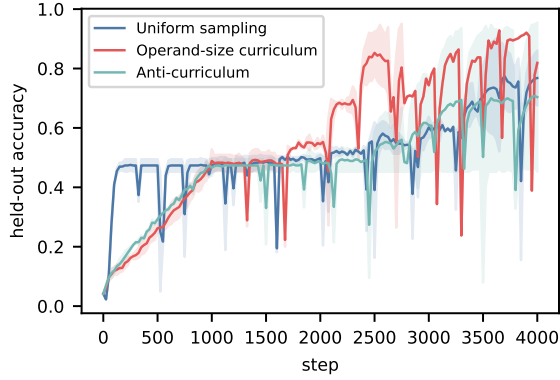


Fig. 2. Held-out accuracy versus step at the main cell, mean over the two extra seeds 5 and 6 (shaded: range). Runs stop once they reach 95%, so late points average fewer runs. The vertical axis is held-out accuracy.

TABLE IV

Weight decay $\lambda \times$ training fraction f grid. Each entry is mean steps to 95% (censored at 4000) with the number of seeds that reached 95% in brackets; n is the seed count. The (1,0.5) row is the main group.

λ	f	n	Unif.	Curr.	Anti
0	0.4	3	4000 (0/3)	4000 (0/3)	4000 (0/3)
0	0.5	3	4000 (0/3)	4000 (0/3)	4000 (0/3)
0	0.7	3	4000 (0/3)	4000 (0/3)	4000 (0/3)
1	0.4	3	4000 (0/3)	4000 (0/3)	4000 (0/3)
1	0.5	5	2945 (3/5)	3955 (1/5)	3990 (1/5)
1	0.7	3	1817 (3/3)	1733 (3/3)	2575 (3/3)
3	0.4	3	4000 (0/3)	4000 (0/3)	4000 (0/3)
3	0.5	3	4000 (0/3)	4000 (0/3)	3358 (1/3)
3	0.7	3	1142 (3/3)	1267 (3/3)	2358 (2/3)

slower at $\lambda = 3$ (1267 vs. 1142). The sign flips, so the registered refutation rule (“same sign in every cell where both reach 95%”) is not met, but both differences are much smaller than the seed spread with 3 seeds, so we do not claim an interaction. In the remaining cells the curriculum is never the only system to generalise; in (3,0.5) only the anti-curriculum did (1 of 3 seeds). Training fraction and weight decay themselves matter far more than the order: nothing generalised at $f = 0.4$ for any λ , and only the cells with $\lambda \in \{1, 3\}$ and $f \geq 0.5$ produced successes.

VI. Ablations and Analysis

Curriculum length and ordering control. At the main cell we varied the curriculum length T_c and replaced the operand-size order by a random order with the same pacing (Table V, Fig. 3a; 5 seeds each; $T_c = 1000$ is the main-group row).

Shorter curricula are better: with $T_c = 250$ the curriculum reaches 95% in 5 of 5 seeds in 2905 steps on average, close to uniform (2945, 3 of 5), and the means degrade monotonically as T_c grows, down to 0 of 5 reached at $T_c = 4000$. The two longest curricula are confounded with the fixed 4000-step budget: the pool is not full

TABLE V

Pacing sweep and shuffled-order control at the main cell (5 seeds). final acc is held-out accuracy when the run stopped (at the 95% crossing or at 4000 steps).

Variant	steps_to_95	reached	final acc
curriculum, $T_c = 250$	2905 $\pm$ 727	5/5	0.96
curriculum, $T_c = 500$	3320 $\pm$ 893	3/5	0.84
curriculum, $T_c = 1000$	3955 $\pm$ 101	1/5	0.75
curriculum, $T_c = 2000$	3990 $\pm$ 22	1/5	0.59
curriculum, $T_c = 4000$	4000 $\pm$ 0	0/5	0.54
shuffled order, $T_c = 1000$	4000 $\pm$ 0	0/5	0.49
uniform (main group)	2945 $\pm$ 1020	3/5	0.86

until step T_c and the training set is fitted only at about 1875 ($T_c = 2000$) and 3700 ($T_c = 4000$) steps, so little time is left for generalisation and part of the 0/5 and 1/5 outcomes is by construction. The clean contrast is $T_c = 250$ against $T_c = 1000$. The shuffled-order run with $T_c = 1000$ reached 95% in 0 of 5 seeds against 1 of 5 for the operand-size order. With 5 seeds this one-seed difference is not evidence for or against a benefit of ordering by operand size; what the sweep does show is that, within the curriculum family, a slower pool growth reached 95% less often (5 of 5 at $T_c = 250$ against 1 of 5 at $T_c = 1000$). Relative to uniform sampling, however, no pacing or ordering was better, and the 7-seed check above does not establish that any of them is worse. A plausible reason (we did not isolate it) is that a small early pool is memorised quickly and the model then needs the remaining examples before it can find a solution that works for all pairs, so delaying full exposure delays generalisation; the training-accuracy column (about 960 steps to fit the training set under both orderings vs. about 125 for uniform) is consistent with this but does not prove it. We did not test an intermediate-sized operand-size curriculum with a different difficulty measure.

Weight decay and training fraction. The grid (Table IV) is the second sensitivity study. Reading it against H3 and H4: generalisation within 4000 steps needed both a nonzero weight decay and a large enough training fraction, and in this region the ordering made at most a small difference compared with these two knobs. At $f = 0.7$ the anti-curriculum was slower than uniform at $\lambda = 1$ (2575 vs. 1817) and at $\lambda = 3$ (2358 vs. 1142, with 2 of 3 seeds reaching 95%), which is the clearest ordering effect in the grid, but with 3 seeds it remains suggestive.

VII. Limitations

- Scale. One modulus (23), one one-layer architecture, one learning rate (3×10^{-3} , chosen on uniform sampling with an unreported pilot seed, which favours uniform), 4000 steps. A different modulus, learning rate, depth or step budget could change the ordering of the systems.
- Censoring and power. Most grid cells have all runs censored at 4000 steps, so they carry no information

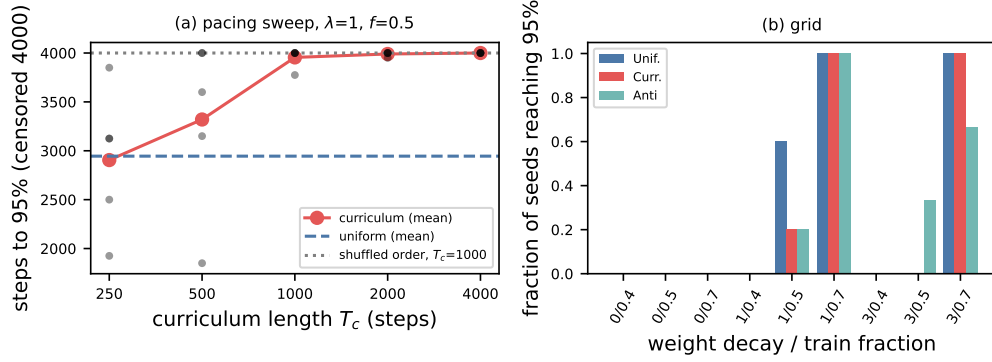


Fig. 3. (a) Steps to 95% against curriculum length T_c (black dots: seeds; lines: means, uniform and shuffled-order references). (b) Fraction of seeds reaching 95% in each weight-decay/training-fraction cell (3 seeds; 5 at 1/0.5).

about relative speed. The main cell has 5 seeds and the grid 3; the test registered for H1 is not significant (Table II), so the data neither establish nor exclude a difference. Means of censored values are lower bounds; we reported the fraction reached alongside.

- Grokking only partly reached. Within the budget, delayed generalisation was seen at the main cell for uniform sampling and in the cells with $\lambda \geq 1$ and $f \geq 0.5$; at $\lambda = 0$ and at $f = 0.4$ it was not reached by any run, and we cannot say if longer runs would.
- Training instability. Held-out accuracy curves show sharp transient drops (Fig. 2), and one uniform run at the main cell ended with a training accuracy below 0.9. A smaller learning rate with longer training would be a cleaner test than ours.
- Narrow curriculum family. A single difficulty measure ($\max(a, b)$), a single linear pacing function and rank-based pools; other measures (e.g. sum size or residue structure) and exposure-matched baselines were not tried. The explanation for the pacing result is a hypothesis, not tested by an isolating run.
- Reimplementation. The uniform baseline and the model are our own code; no result was compared to published numbers, and no mechanistic analysis of the learned solution was done.
- Held-out definition. Held-out accuracy is on the complement of a random split; both ordering systems see exactly the same training pairs as uniform, only their timing differs.

VIII. Conclusion

In a one-layer transformer on $(a + b) \bmod 23$, an operand-size curriculum did not reach 95% held-out accuracy faster than uniform sampling at the main cell (3955 vs. 2945 mean censored steps; 1 vs. 3 of 5 seeds), and the anti-curriculum was no better. The difference is not significant with 5 seeds, and two further seeds run afterwards reversed its direction (pooled 7 seeds: 3789 vs. 3246 steps, 2 vs. 3 of 7 reaching 95%, not significant), so no ordering showed a benefit and we do not establish a loss.

What the pacing sweep does support is an effect within the curriculum family ($T_c = 250$ against $T_c = 1000$), which we hypothesise, without an isolating run, comes from delayed full exposure of the training set rather than from operand size. Weight decay and training fraction dominated: no run generalised at $\lambda = 0$ or $f = 0.4$ within 4000 steps, and the curriculum-versus-uniform sign flipped between the two cells where both always succeeded, within noise. A full study would add more moduli, longer budgets with smaller learning rates, more seeds, and other difficulty measures.

References

- [1] A. Power, Y. Burda, H. Edwards, I. Babuschkin, and V. Misra, “Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets,” arXiv preprint arXiv:2201.02177, 2022. [Online]. Available: <https://arxiv.org/abs/2201.02177>
- [2] N. Nanda, L. Chan, T. Lieberum, J. Smith, and J. Steinhardt, “Progress measures for grokking via mechanistic interpretability,” arXiv preprint arXiv:2301.05217, 2023. [Online]. Available: <https://arxiv.org/abs/2301.05217>
- [3] A. Gromov, “Grokking modular arithmetic,” arXiv preprint arXiv:2301.02679, 2023. [Online]. Available: <https://arxiv.org/abs/2301.02679>
- [4] Z. Liu, O. Kitouni, N. Nolte, E. J. Michaud, M. Tegmark, and M. Williams, “Towards Understanding Grokking: An Effective Theory of Representation Learning,” arXiv preprint arXiv:2205.10343, 2022. [Online]. Available: <https://arxiv.org/abs/2205.10343>
- [5] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, “Curriculum learning,” Proceedings of the 26th Annual International Conference on Machine Learning, 2009. [Online]. Available: <https://doi.org/10.1145/1553374.1553380>
- [6] X. Wang, Y. Chen, and W. Zhu, “A Survey on Curriculum Learning,” arXiv preprint arXiv:2010.13166, 2020. [Online]. Available: <https://arxiv.org/abs/2010.13166>
- [7] Z. Liu, E. J. Michaud, and M. Tegmark, “Omnigrok: Grokking Beyond Algorithmic Data,” arXiv preprint arXiv:2210.01117, 2022. [Online]. Available: <https://arxiv.org/abs/2210.01117>
- [8] M. A. Mohamadi, Z. Li, L. Wu, and D. J. Sutherland, “Why Do You Grok? A Theoretical Analysis of Grokking Modular Addition,” arXiv preprint arXiv:2407.12332, 2024. [Online]. Available: <https://arxiv.org/abs/2407.12332>
- [9] S. Golechha, “Progress Measures for Grokking on Real-world Tasks,” arXiv preprint arXiv:2405.12755, 2024. [Online]. Available: <https://arxiv.org/abs/2405.12755>
- [10] I. Loshchilov and F. Hutter, “Decoupled Weight Decay Regularization,” arXiv preprint arXiv:1711.05101, 2017. [Online]. Available: <https://arxiv.org/abs/1711.05101>

- [11] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention Is All You Need,” arXiv preprint arXiv:1706.03762, 2017. [Online]. Available: <https://arxiv.org/abs/1706.03762>